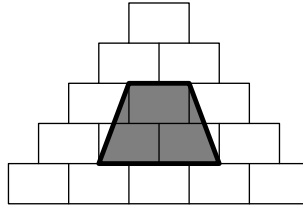
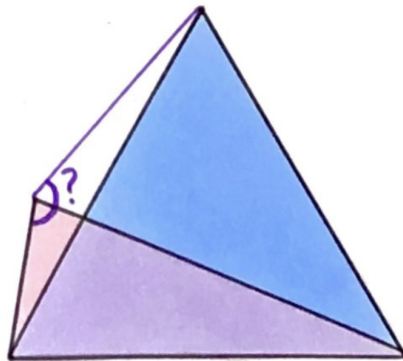


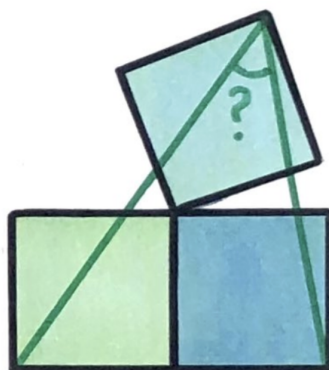
1. John wants to travel from  $(0, 0)$  to  $(5, 2)$  while touching the river  $y = 6$ . Find, with reason, the minimum distance John must travel?
2. Five layers of 15 identical rectangles are placed, as shown. If the area of the shaded region is 12, what is the area of the entire figure?



3. Let  $ABCD$  be a rectangle such that  $AB = 6$  and  $BC = 8$ . Let  $X$  and  $Y$  be points on segments  $BC$  and  $CD$  respectively, and let  $Z$  be the midpoint of segment  $DA$ . Find the minimum value of  $AX + XY + YZ$ .
4. A blue equilateral triangle overlaps a pink isosceles triangle. What's the angle?



5. All three squares are the same size. What's the angle?



6. Let  $X$  be a point inside equilateral triangle  $ABC$  such that  $AX = \sqrt{2}$ ,  $BX = 3$ , and  $CX = \sqrt{5}$ . Find the measure of  $\angle AXB$ , in degrees.
7. Let  $ABCD$  be a concave quadrilateral with  $AB = 12$ ,  $BC = 16$ ,  $AD = CD = 26$ , and  $\angle ABC = 270^\circ$ . Find the area of  $ABCD$ .

8. Consider a right trapezoid with side lengths 10, 30, and 40 inscribed in an isosceles right triangle, as shown below. Determine the area of the isosceles right triangle.

